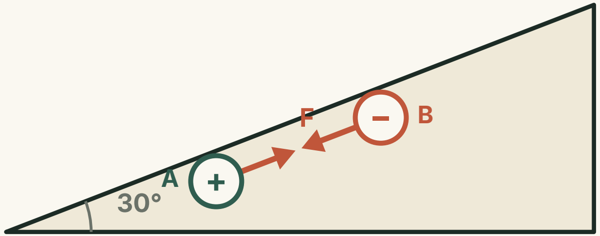
ELECTROSTATICS

The Coulomb force: balance & acceleration

When charges also feel gravity, tension or a push along a slope.



ESSENTIAL QUESTION

When a charged object sits still or speeds up, is this new physics — or just mechanics with one extra force?

LESSON MAP

One idea, built in four moves.

01

FRAME

The Coulomb force is just one more force on the diagram.

02

BALANCE

At rest, every force cancels.

03

THREE CHARGES

Free charges in a line settle by simple rules.

04

ACCELERATE

Speeding up together under a net force.

THE WHOLE TRICK

An electric problem, solved with mechanics: draw the forces, name the state, then apply balance or ma.

RECALL

The Coulomb force is just another force.

It acts along the line joining the charges — push apart if alike, pull together if opposite.

COULOMB'S LAW

$$F = k\,\frac{q_1 q_2}{r^2}$$

k is the electrostatic constant, q the charge magnitudes, r their separation. Equal and opposite on the two charges — Newton's third law still holds.

METHOD

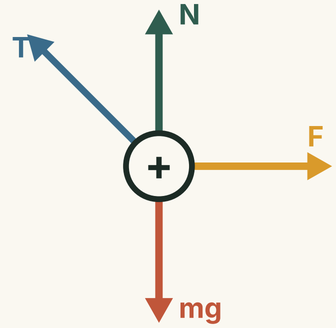
An electric problem, a mechanics method.

Charged bodies obey the same three steps as any force problem.

1. **Draw the free-body diagram**
   Mark gravity, normal, tension, friction — and the Coulomb force, like any other.

2. **Name the state**
   At rest or constant velocity → equilibrium. Speeding up → accelerating.

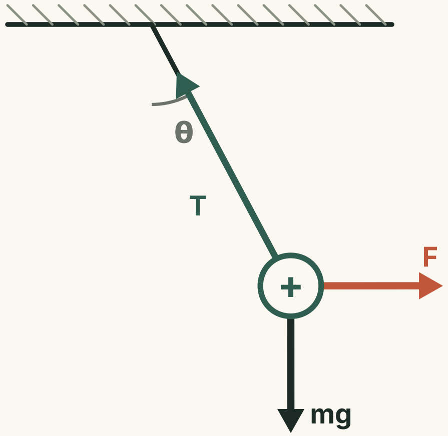
3. **Apply the law**
   Equilibrium → balance; accelerating → ma. Then solve.



BALANCE

At rest, three forces close into a triangle.

A charged ball hangs by a string near a fixed charge; the string leans at angle θ.



RESOLVE ALONG AND ACROSS THE STRING

VERTICAL & HORIZONTAL

$$T\cos\theta = mg \quad T\sin\theta = F$$

THE LEAN ANGLE ALONE FIXES THE FORCE

$$F = mg\tan\theta$$

Tension follows: $T = mg / \cos\theta$

Three free charges settle by four rules.

If all three rest in equilibrium on their own, their layout is forced.

Collinear

All three lie on one straight line.

Signs alternate

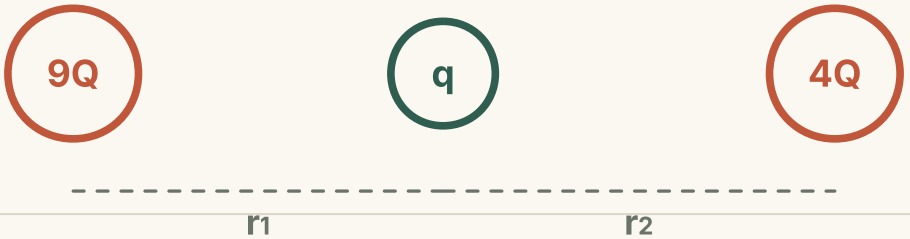
Two like charges flank one opposite charge.

Smallest inside

The middle charge has the least charge.

Nearer the smaller

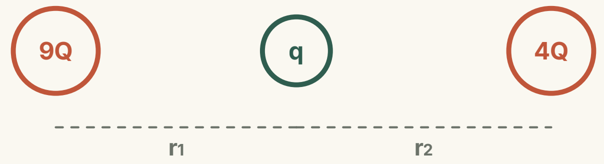
The middle sits closer to the smaller outer charge.



DERIVE

The middle charge sits where the two pulls cancel.

Balance the forces on the inner charge — its sign and size drop out.



EQUAL AND OPPOSITE PULLS

$$\frac{k(9Q)q}{r_1^2} = \frac{k(4Q)q}{r_2^2}$$

SOLVE FOR THE POSITION

$$\frac{r_1}{r_2} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

Closer to the smaller charge, as promised. For all three to balance, the inner charge must be **negative**.

ACCELERATE

Moving together? Choose system or isolation.

Two methods, one law — pick the body that makes the Coulomb force vanish or appear.

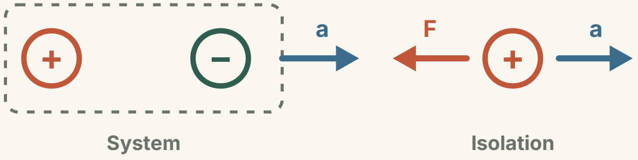
SYSTEM METHOD — INTERNAL FORCES CANCEL

$$\sum F_{\text{ext}} = (m_1 + m_2)\, a$$

ISOLATION METHOD — CUT ONE BODY FREE

$$\sum F = ma$$

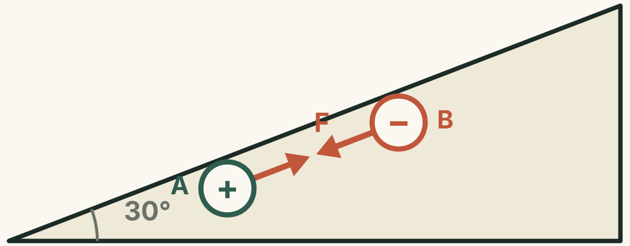
Now the Coulomb force does appear.



APPLY

Two charges, one push, up a frictionless incline.

A above, B below, each mass m, kept a distance r apart by attraction; a push drives them up a 30° slope.



ISOLATE B

The attraction pulls it up-slope.

ISOLATE A

The Coulomb pull drags it down-slope.

APPLY TO EACH BODY

$$\sum F = ma$$

CHECK

Three traps to avoid.

Most mistakes here are mechanics mistakes, not electricity ones.

01 · Don’t drop a force

Gravity and the normal force are still there. The Coulomb force is added, not a replacement.

02 · Pick the right body

Use the system for external forces; isolate one body to find an internal force or shared a.

03 · Mind the direction

Like charges repel, opposite attract. Get the arrow right before resolving.

ALWAYS REMEMBER Same charges repel, opposite attract — always along the joining line.

SUMMARY

Treat the Coulomb force like any other force on the free-body diagram.

THEN APPLY ONE OF TWO LAWS

$$\sum \vec{F} = 0 \quad \text{or} \quad \sum \vec{F} = m\vec{a}$$

Decide whether the body is **balanced** or **accelerating**, then solve.

Electric problem, mechanics method.